In [10]:
```
# %pip install pandas numpy matplotlib seaborn scikit-learn scipy openpyxl p
```

In [11]:
```
# %pip install opencv-python
```

In [ ]:
```
# 1)
# Write a program non-recursive and recursive program to calculate Fibonacci
# numbers and analyze their time and space complexity.
# 02
# 2)
# Write a program to implement Huffman Encoding using a greedy strategy
# 01
# 3)
# Write a program to solve a fractional Knapsack problem using a greedy
# method. 01
# 4)
# Write a program to solve a 0-1 Knapsack problem using dynamic programming
# or branch and bound strategy.
# 02
# 5)
# Design n-Queens matrix having first Queen placed. Use backtracking to plac
# remaining Queens to generate the final n-queen's matrix. 01
# 6)
# Write a program to implement dictionary using hashing. (Use separate
# # chaining)
```

In [155…
```
def fibanacci_non_recursive(n):
    a,b = 0,1

    for i in range(n):
        print(a, end=" ")
        a,b = b, a+b


def fibanacci_recursive(n):
    if n<=1:
        return n
    return fibanacci_recursive(n-1) + fibanacci_recursive(n-2)


n= int(input("Enter the no. of terms:"))

if n <0:
    print("Enter positive integer")
else:
    print("non-Recurssive fibanacci series:")
    fibanacci_non_recursive(n)

    print("\nrecurssive function:")

    for i in range(n):
        print(fibanacci_recursive(i), end=" ")
```

```
non-Recurssive fibanacci series:
0 1 1 2 3 5
recurssive function:
0 1 1 2 3 5
```

In [ ]: